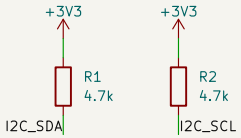
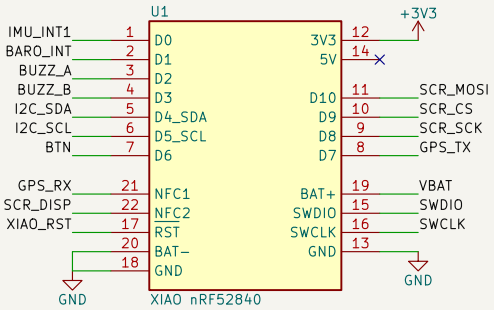
Board notes: keep U2 away from the XIAO antenna end and regulator, under a foam cover with a vent.
Put U3 near the board centre, away from screws. The XIAO antenna end overhangs the carrier or sits over a copper keepout.

I2C pull-ups

To 3V3, i2c1 at 400 kHz.



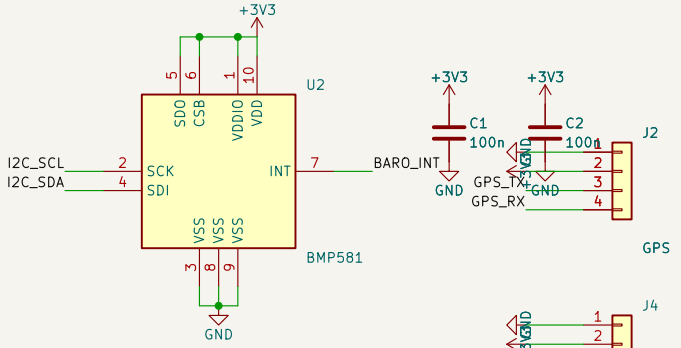
U1 Seeed XIAO nRF52840 (non-Sense)



Onboard: 3.3 V LDO (3V3 pin), BQ25101 LiPo charger, USB-C, UF2 bootloader.
Internal, no carrier connection: P0.31 VBAT_SENSE, P0.14 VBAT_SENSE_EN,
P0.17 CHG_STAT, P0.13 CHG_ISET. Spares: D7–D10 (SPI/UART), NFC1/NFC2.

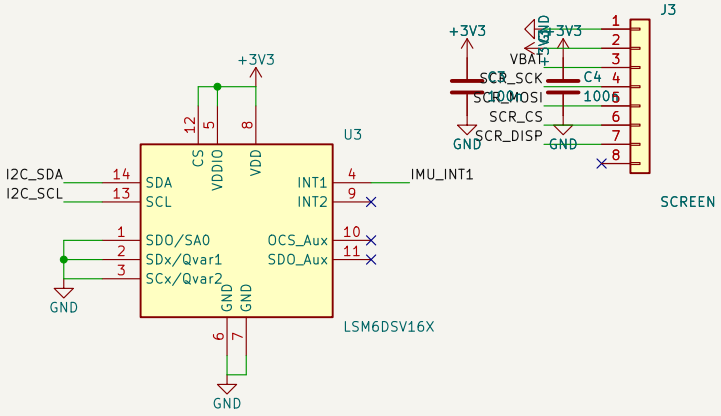
U2 Barometer (BMP581)

I2C address 0x47: SDO and CSB tied to 3V3.
C1 at VDDIO (pin 1), C2 at VDD (pin 10), each close to its pin.



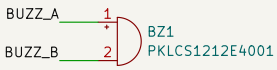
U3 IMU (LSM6DSV16X)

I2C address 0x6A: SDO/SA0 to GND, CS to 3V3.
C3 at VDDIO (pin 5), C4 at VDD (pin 8). Qvar pins 2–3 to GND.
INT2 and aux SPI (pins 10–11) unused.



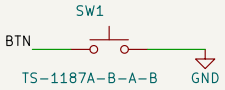
BZ1 Piezo beeper

Driven differentially by PWM1 ch0/ch1, about 6.6 Vpp.



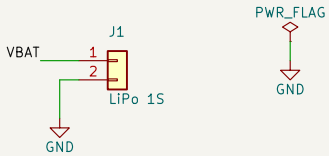
SW1 User / power button

To GND, internal pull-up. Wakes from System OFF.



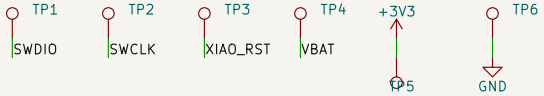
J1 1S LiPo, about 150 mAh

Pin 1 = +. Check the cell lead polarity before plugging in.
Charged by the XIAO through its BAT+ / BAT- underside pads.



Debug pads

SWD for a J-Link or Pico debug probe, wired to the XIAO underside pads.



Pin map: docs/airio-parts-and-pinmap.md
XIAO nRF52840 carrier: BMP581 baro, LSM6DSV16X IMU, piezo, button, LiPo

Flario

Sheet: /
File: airio.kicad_sch

Title: airio

Size: A4	Date: 2026-10-24	**Rev: 0.1**
KiCad E.D.A. 10.0.7		Id: 1/1